

Spotify Listening Behavior Analysis Using Audio Features and Machine Learning

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Abstract—This project analyzes a personal Spotify listening history enriched with Spotify audio features to study music preferences, listening patterns, and repeated listening behavior. Milestone 1 focused on exploratory data analysis and non-parametric hypothesis testing. Milestone 2 extends the project with a reproducible machine learning workflow that predicts whether a track is frequently played using audio features and selected Spotify metadata. The workflow uses leakage-aware preprocessing, stratified train/test splitting, cross-validated hyperparameter tuning, and model comparison across traditional classifiers. The best held-out F1-score was obtained by Random Forest ($F1 = 0.400$), which achieved high precision but low recall because the frequently played class was rare.

I. INTRODUCTION / MOTIVATION

Music listening behavior can reflect personal preferences, mood, routines, and repeated engagement with specific artists or tracks. Spotify listening history provides a rich opportunity to analyze these patterns using both behavioral data and audio features. This project investigates whether Spotify audio features such as energy, valence, danceability, tempo, acousticness, and loudness can help explain listening patterns and predict whether a track is repeatedly played.

The project is designed as an academically honest and reproducible DSA 210 data science workflow. The raw listening history is private and therefore excluded from GitHub. Code, documentation, figures, and report scaffolding are included so the analysis can be reproduced locally by placing the private JSON file in the project root.

II. DATA SOURCE

The dataset consists of personal Spotify Extended Streaming History records enriched with Spotify Web API audio features. The private data file is named `gulcan_spotify_formatted.json`. The formatted dataset contains 1,558 play events, 1,535 unique track names, and 921 unique artists covering April 2025 to April 2026.

III. DATA COLLECTION

The listening records include track name, artist name, playback timestamp, and audio features. The raw JSON file is excluded from version control because it contains personal listening history. The repository includes notebooks and generated figures, but not the private raw data.

IV. DATA PREPROCESSING

The Milestone 2 notebook implements a preprocessing workflow using scikit-learn pipelines. Numeric variables are checked for missing values and skewness. Strongly skewed positive numeric variables are selected from the training split only and transformed using `log1p`. Numeric variables are imputed and scaled inside the model pipeline. Binary categorical variables such as Spotify mode are imputed and passed as binary values. Non-binary categorical variables such as Spotify key are one-hot encoded.

Track and artist text fields are detected and normalized through lowercase conversion, punctuation removal, and stop-word removal. However, these identity-like text fields are excluded from the primary predictive model because they may encourage memorization rather than a generalizable claim about audio features. Date/time variables are engineered for analysis but excluded from the main model input to reduce leakage from listening opportunity.

V. EXPLORATORY DATA ANALYSIS

Milestone 1 visualizations show distributions of Spotify audio features, feature correlations, top artists and tracks, monthly activity, hourly activity, valence over time, and energy-valence relationships.

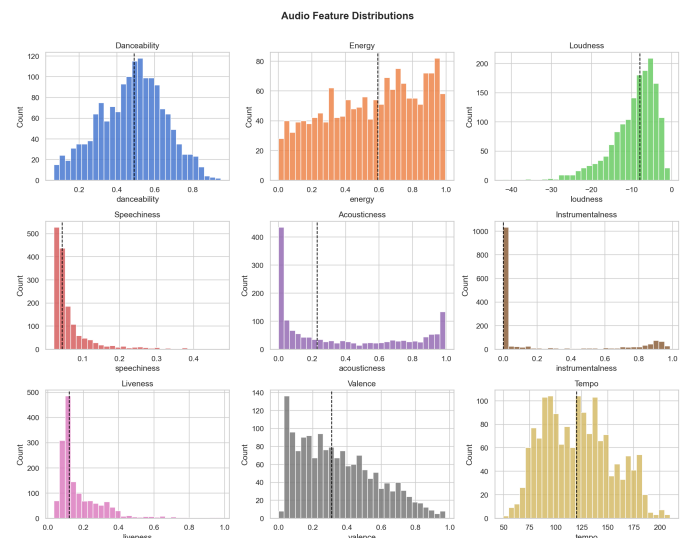


Fig. 1. Distribution of Spotify audio features in the listening history.

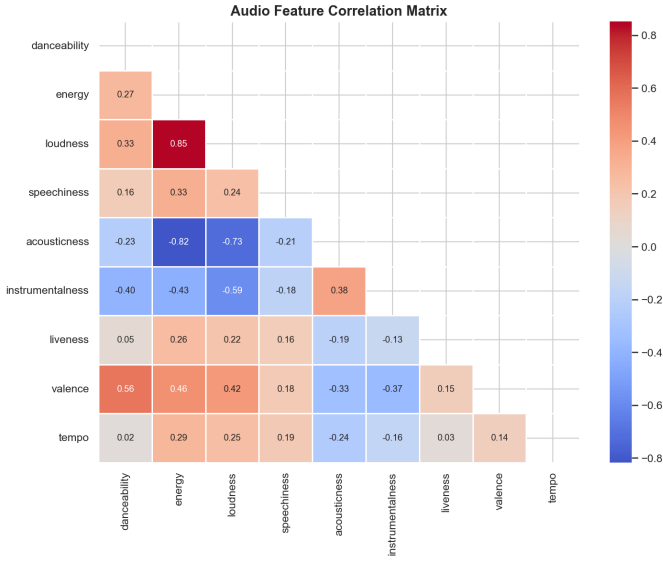


Fig. 2. Correlation heatmap for Spotify audio features.

VI. HYPOTHESIS TESTING

Milestone 1 tested four hypotheses using non-parametric methods where appropriate. The analysis found no statistically significant evidence that high-energy songs were played more frequently than low-energy songs, and no significant seasonal difference in valence between summer and winter. The analysis found evidence that acoustic songs had lower tempo than non-acoustic songs and that energy and danceability were positively correlated.

TABLE I
MILESTONE 1 HYPOTHESIS TEST SUMMARY.

Hypothesis	Test	p-value	Result
High-energy songs are played more	Mann-Whitney U	0.8707	Not rejected
Higher valence in summer	Mann-Whitney U	0.8250	Not rejected
Acoustic songs have lower tempo	Mann-Whitney U	< 0.0001	Rejected
Energy and danceability are positively correlated	Spearman rho	< 0.0001	Rejected

VII. MACHINE LEARNING METHODS

Milestone 2 frames the problem as binary classification. A track is labeled as frequently played if its play count is at least two. The primary feature set includes audio features and selected Spotify metadata while excluding leakage-prone fields such as play count, first play time, last play time, track name, artist name, and text metadata.

The notebook evaluates Logistic Regression, Gaussian Naive Bayes, Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and AdaBoost. Hyperparameters are tuned using cross-validation with a stratified split. If class imbalance is detected, F1-score is used as the primary tuning metric; otherwise accuracy is used.

VIII. RESULTS

The track-level modeling table contains 1,535 tracks. The target is highly imbalanced: 1,513 tracks were played once, while only 22 tracks were played at least twice. Because of this imbalance, accuracy alone is not a reliable success metric. F1-score, recall, balanced accuracy, and ROC-AUC were used to interpret the models more carefully.

TABLE II
MILESTONE 2 MODEL COMPARISON ON THE HELD-OUT TEST SET.

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
Random Forest	0.990	1.000	0.250	0.400	0.748
AdaBoost	0.987	0.500	0.250	0.333	0.808
Decision Tree	0.984	0.333	0.250	0.286	0.622
SVM	0.980	0.250	0.250	0.250	0.757
Logistic Reg.	0.648	0.019	0.500	0.036	0.604
Naive Bayes	0.277	0.018	1.000	0.035	0.752
Gradient Boost.	0.987	0.000	0.000	0.000	0.674

Random Forest was selected as the best balanced model by held-out F1-score. Its tuned configuration used class weighting, 100 trees, maximum depth 10, and minimum leaf size 5. It predicted very few positive cases, but the positive cases it predicted were correct, producing precision of 1.000 and recall of 0.250. This indicates that the model is conservative: it can identify some frequently played tracks, but it misses most of them.



Fig. 3. Target distribution for the frequently played binary classification task.

IX. DISCUSSION

The results suggest that Spotify audio features contain some signal for repeated listening behavior, but the signal is limited. The best model reached high test accuracy, but this is partly caused by the extreme class imbalance. Random Forest achieved the strongest F1-score because it avoided false positives while finding one quarter of the frequently played tracks in the held-out split. AdaBoost produced the highest ROC-AUC among the tested models, but its F1-score was lower than Random Forest because its positive-class precision and recall tradeoff was weaker.

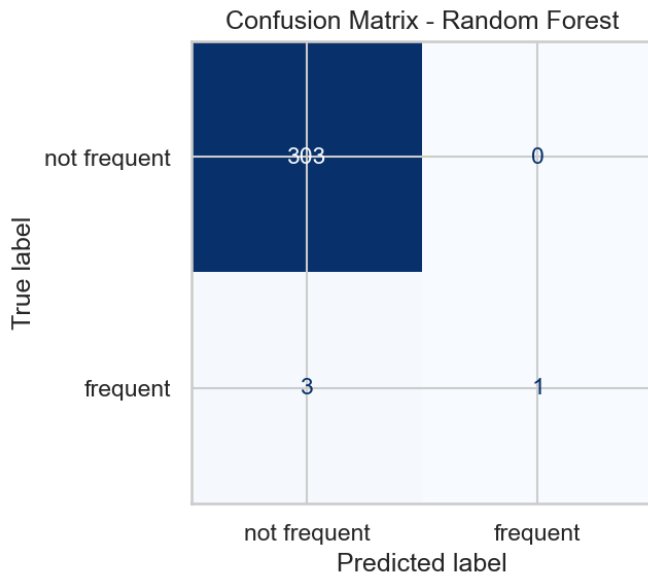


Fig. 4. Confusion matrix for the tuned Random Forest classifier.

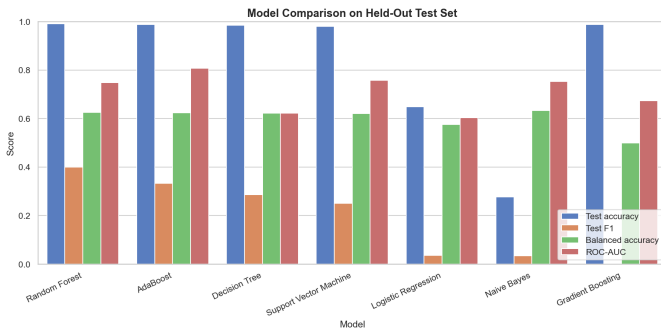


Fig. 5. Comparison of tuned machine learning models on the held-out test set.

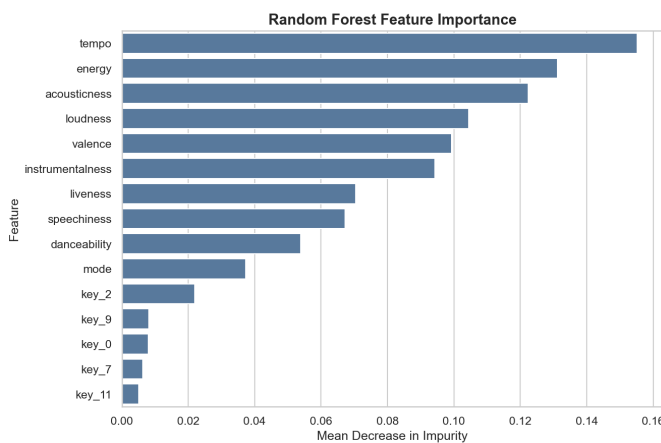


Fig. 6. Random Forest feature importance for the primary audio-feature model.

Naive Bayes reached recall of 1.000, but its precision was only 0.018 and its accuracy was low, meaning it labeled too many tracks as frequently played. Gradient Boosting achieved high accuracy but did not identify any positive examples, which makes it unsuitable for this imbalanced target despite its apparent accuracy. Overall, the modeling results should be interpreted cautiously: audio features alone are not enough to fully explain repeated listening, but they provide a reproducible baseline for comparing more context-aware future models.

X. LIMITATIONS

The project uses one person's listening history, so findings should not be generalized to broader Spotify users. The target variable is based on repeated listening within the available history and may be affected by time window length, playlist behavior, mood, seasonality, and external context. Audio features alone may not capture social, lyrical, cultural, or situational reasons for repeated listening.

XI. FUTURE WORK

Future work could evaluate temporal validation, include additional context features, compare audio-only models with carefully controlled text or artist-level baselines, and analyze model stability across different target thresholds such as three or more plays.

XII. ETHICAL CONSIDERATIONS

The raw data contains personal listening history and is therefore excluded from the public repository. The analysis should avoid overclaiming psychological or emotional conclusions from audio features alone. Any sharing of the project should preserve privacy and avoid exposing personally sensitive listening records.

XIII. AI ASSISTANCE DISCLOSURE

AI assistance was used to improve project organization, preprocessing design, machine learning pipeline structure, repository documentation planning, and this LaTeX scaffold. The assistance did not fabricate data or model results. The Milestone 2 notebook was rerun with the private JSON file before reporting the model results in this document. This section is the project's AI assistance disclosure.

XIV. REFERENCES

REFERENCES

- [1] Spotify for Developers, "Web API Reference," <https://developer.spotify.com/documentation/web-api>.
- [2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [3] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.